

Electrotactile Assistive Tools to Support Blind and Low Vision People's Data Comprehension

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Abstract

Being able to access and interpret data is a critical skill in today's world, but it remains challenging for blind and low vision (BLV) people. While the accessibility community has explored various solutions to make data more accessible, little is known about BLV people's daily data analysis (DDA) workflows and how haptic assistive tools, especially electrotactile devices, can support their data comprehension. To address these gaps, my dissertation investigates how BLV people engage in DDA and develops electrotactile assistive tools to support common scenarios such as chart and map understanding, and trip preview based on the study results. Through a series of user studies, I contribute empirical insights into five key strategies used by BLV people for DDA, and two electrotactile systems, providing evidence that these systems are effective when optimized for different scenarios. I plan to conduct further research on optimizing electrotactile feedback through a deeper understanding of its perceptual mechanisms, supported by AI-driven adaptation and personalization.

CCS Concepts

• **Human-centered computing** → **Empirical studies in accessibility**; **Haptic devices**.

Keywords

blind and low vision, daily data analysis, electrotactile feedback, charts understanding, trip planning

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1 Introduction

Access to data is essential for people in both their daily work and personal lives. However, BLV people face challenges in accessing and comprehending data, particularly when interpreting complex and dynamic information such as charts, figures, and maps [36]. To

address these challenges, researchers have developed various assistive technologies, including printed Braille and refreshable Braille displays [15, 18], screen readers [27, 35], sonification techniques [17, 33], and haptic devices [9, 15].

Despite these advancements, such solutions are often not well integrated into BLV users' daily workflows, particularly in tasks involving unfamiliar data representations, chart comprehension, or spatial information understanding during independent travel [36]. Moreover, many haptic assistive tools, such as robotic arms and refreshable Braille displays, are bulky and heavy, limiting their flexibility and usability across diverse contexts such as navigation, education, and work [19]. Furthermore, although prior research on electrotactile feedback has explored its applications, the underlying perceptual mechanisms remain underexplored. This gap hinders its optimization for future assistive applications.

To address these gaps, my dissertation research follows a systematic approach: (1) understanding BLV people's daily data access and analysis workflows; (2) designing and developing tailored electrotactile devices to support BLV people's chart comprehension; (3) improving the electrotactile system to assist BLV people's map understanding and trip preview, and (4) finally, leveraging AI-assisted techniques to deepen our understanding of electrotactile perception mechanisms and optimize feedback parameters accordingly.

In sum, my dissertation aims to advance haptic solutions for BLV people's data access through a user-centered and systematic approach, making the following contributions:

- I identified the current practices and challenges of BLV people's daily data analysis (DDA) [20];
- I designed and evaluated an electrotactile assistive tool to support BLV people with understanding commonly-used charts [19];
- I developed an AI-assisted electrotactile assistive tool for BLV people's map understanding and trip preview [*Under Review*].
- Finally, I propose future research on AI-assisted electrotactile feedback optimization through psychophysical mechanism understanding and its application in more scenarios.

2 How Can BLV People Conduct Daily Data Analysis (ASSETS'23)

We first explored how BLV people perform DDA in their daily life and work. DDA refers to common data-related tasks people encounter in routine situations, such as splitting expenses among friends, tracking changes in stock portfolios, or calculating average scores. The term "daily" highlights the frequent and practical nature of these activities.

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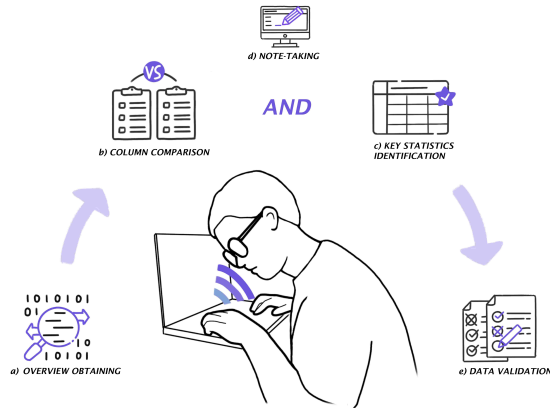


Figure 1: Our research revealed five key approaches to performing DDA among BLV people: a) Overview Obtaining; b) Column Comparison; c) Key Statistics Identification; d) Note-Taking; e) Data Validation.

Prior research has examined BLV people’s data access practices across various platforms and contexts, providing accessibility recommendations and design guidelines. These studies have addressed web browsing and social media [8, 36], smartphone usage [3, 28], and online shopping [24, 26]. To enhance data access, researchers have developed assistive technologies such as screen readers [27, 35], sonification methods [17, 33], Braille displays [15, 18], and haptic devices [9, 15]. In addition to addressing access challenges, prior work has explored how BLV people navigate spreadsheets non-visually [6, 21, 38, 39]. However, most of these studies focused on navigation rather than data comprehension. In contrast, DDA involves both navigating the data and making sense of its meaning. How BLV people perform DDA and what challenges they face in doing so remains largely unexplored. To address this gap, I investigated the following research question (RQ): **How do BLV people perform DDA, and what are the associated challenges?**

To answer this question, I conducted a mixed-methods study with 16 BLV participants, using semi-structured interviews and think-aloud sessions. During these sessions, participants completed a series of DDA tasks using screen readers while verbalizing their thought processes. I identified five key approaches that BLV people use during DDA: overview obtaining, column comparison, key statistics identification, note-taking, and data validation, which are shown in Figure 1. I also discovered particularly challenging operations, such as obtaining the quantity, layout, and relationships between spreadsheet elements. These findings point to opportunities for improving assistive technologies to better support efficient and meaningful data comprehension.

3 Electrotactile Feedback for Blind and Low Vision (BLV) People’s Charts Understanding (CHI’24 Honorable Mention)

Based on my findings on how BLV people conduct DDA, my collaborators and I proposed using electrotactile feedback to support

their comprehension of commonly used charts. Graphical information, including graphs, charts, and maps (commonly referred to as infographics), plays a crucial role across various domains such as education, the workplace, navigation, and daily activities [37, 41]. However, the inherently visual nature of these infographics creates accessibility barriers for BLV people [32]. BLV people’s lack of access to such visual information widens the information gap, particularly for frequently used charts such as line charts, scatter-plots, bar charts, and pie charts [1, 22]. Various haptic assistive tools have been developed to address this challenge, including refreshable Braille displays and pin array haptic displays [15, 18, 31], tactile graphics [13, 29], and other haptic devices [9, 11, 14]. However, some of these tools are specialized for certain types of charts [9, 11, 40], while others rely on bulky or expensive mechanical components [14, 34, 40], or only provide static charts without up-to-date information [2, 10, 29]. These limitations underscore the need for a lightweight, conformal, and wearable device to help BLV people understand various commonly used charts.

To address these challenges, we leveraged electrotactile interfaces, which are smaller, lighter, and more flexible, making them suitable even for elders and children [42]. Prior research on electrotactile feedback primarily applied stimulation to the fingertip pad or the backside of the finger [25, 42, 44], potentially hindering users’ interactions with their surroundings (e.g., smartphones, computers) when wearing the device. This limits the practicality of these devices in daily scenarios. We designed an unobtrusive electrotactile system that applies stimulation to the fingertip edge using a two-channel modulated signal to support BLV users’ comprehension of common charts [19]. The device integrates three core modules: signal generation, stimulation selection, and perception control, as shown in Figure 2. Then, we investigated the research question (RQ): **How can our electrotactile device convey positional and directional information on the fingertip edge to support BLV people in understanding commonly used charts?**

We conducted three studies with twelve BLV participants to evaluate the system’s effectiveness. Our results showed: (a) a position recognition accuracy of 0.789 under the optimal stimulation time of 0.5 s (Study 1); (b) the highest average directional recognition error of just 7.98° with optimized intervals of 25% (Study 2); and (c) a minimal trajectory deviation of 0.21 cm in guided chart reproduction tasks (Study 3). Participants were able to successfully replicate line, bar, and pie charts, demonstrating the device’s potential to facilitate low-level data comprehension tasks.

4 Electrotactile Feedback for Trip Preview (IJHCS Under Review)

Building on our prior work using electrotactile feedback to support BLV people’s understanding of charts, we further enhanced the system to support spatial data access in the context of trip previews. Trip preview refers to the process where travelers review key aspects of an upcoming journey to prepare for a smooth, safe, and efficient trip. This process typically involves familiarizing oneself with important map elements (e.g., destinations and landmarks) and travel route conditions (e.g., routes, weather, and traffic) [4, 12]. However, BLV people face substantial challenges when using conventional trip planning tools like Google Maps and

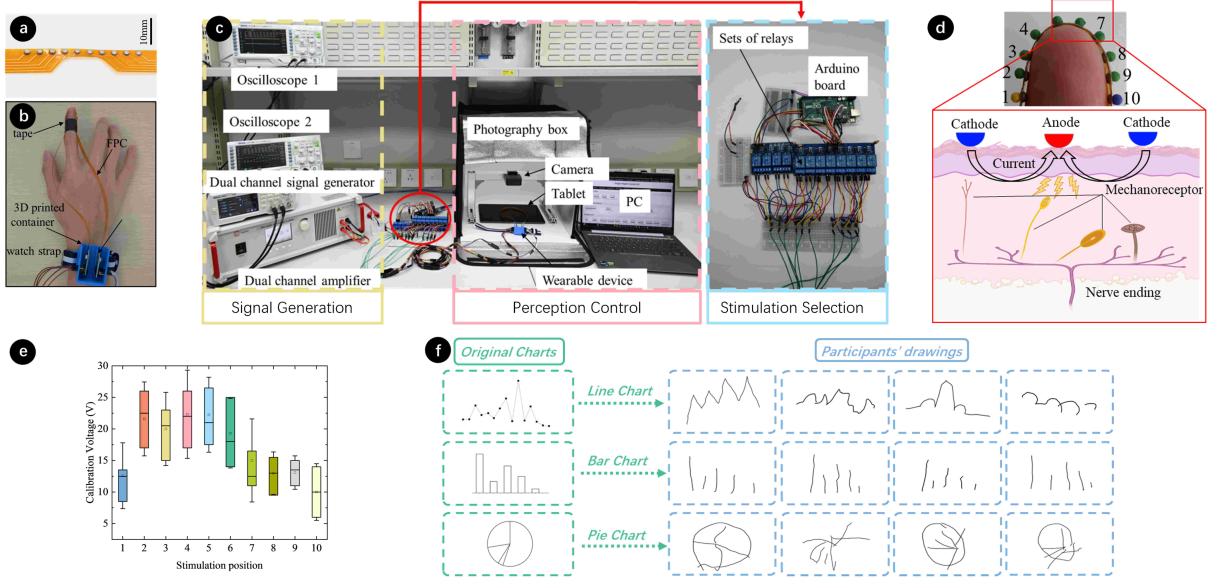


Figure 2: Overview of the unobtrusive electrotactile assistive tool: a) close-up of the flexible printed circuit (FPC); b) user wearing the haptic device; c) complete system showing signal generation, stimulation selection, and perception control; d) front view of the fingertip with the worn FPC (top) and a schematic of electrotactile perception (bottom); e) box plot of calibration voltage results; f) BLV participants' chart replications (green: example charts; blue: replicated charts).

TripAdvisor, as these rely heavily on visual information [7]. This visual dependency often limits BLV users' confidence in traveling to unfamiliar places, affecting their independence and quality of life. Previous work has explored various haptic assistive technologies, such as refreshable Braille displays, tactile maps with 3D icons, and electrotactile devices, to support spatial data comprehension and map overviews [5, 16, 19, 23, 30]. In particular, electrotactile feedback has shown promise for delivering rapid, precise notifications and conveying visual information, such as chart structures, especially when combined with audio feedback [19, 25]. Inspired by the parallels between charts and maps, as both of them contain basic geometric shapes and navigable routes, we extended electrotactile feedback to convey more complex spatial information. Despite its benefits, electrotactile feedback faces key challenges, such as diminished sensation over time and varying sensitivity due to factors like skin-electrode impedance [25, 42]. Conventional calibration methods, including the use of conductive gels, are time-consuming and reduce user independence. BLV users expressed a desire to calibrate and adjust the electrotactile feedback independently, without relying on operators. To address this, we introduced large language models (LLMs) to facilitate personalized calibration and system control. Given the success of LLMs in assisting sighted users with trip previews [43, 45], we designed a system that integrates an LLM-powered Trip Preview Assistant (TPA) with a finger-worn electrotactile device, as shown in Figure 3. The TPA has two primary functions: 1) a voice interface that enables natural interaction with BLV users during trip previews, and 2) an intelligent control agent for dynamically adjusting electrotactile feedback based on user needs. This led us to investigate the following research question:

How can an electrotactile system controlled by an LLM-powered TPA assist BLV people in trip previews?

We conducted a technical evaluation and a three-part user study with twelve BLV participants to answer this research question. In the technical evaluation, we confirmed that the LLM could effectively personalize electrotactile feedback in response to user input. In Part 1 of the user study, participants employed an association memory strategy to efficiently learn and recall electrotactile patterns, achieving recognition accuracies of 98.3% for map elements and 100% for route conditions. In Part 2, participants used the system to preview landmarks, spaces, and routes on maps, achieving an average accuracy of 95% across all tasks. They effectively personalized the electrotactile feedback by instructing the TPA to adjust stimulation intensity and frequency. However, some participants encountered occasional LLM hallucinations, where the TPA generated non-existent information, posing challenges for validation. In Part 3, participants used the electrotactile feedback combined with the TPA to plan and follow desired routes. They demonstrated precise route-following ability, with a low root mean squared (RMS) distance deviation of 0.17 cm (SD = 0.08 cm), indicating accurate adherence to the intended trajectory.

5 Conclusion and Future Work

In conclusion, my dissertation research deepens the understanding of how BLV people interact with data and proposes novel interaction techniques that integrate electrotactile feedback with AI-driven assistance. By addressing key scenarios such as DDA, chart comprehension, and trip preview, my work moves beyond isolated assistive

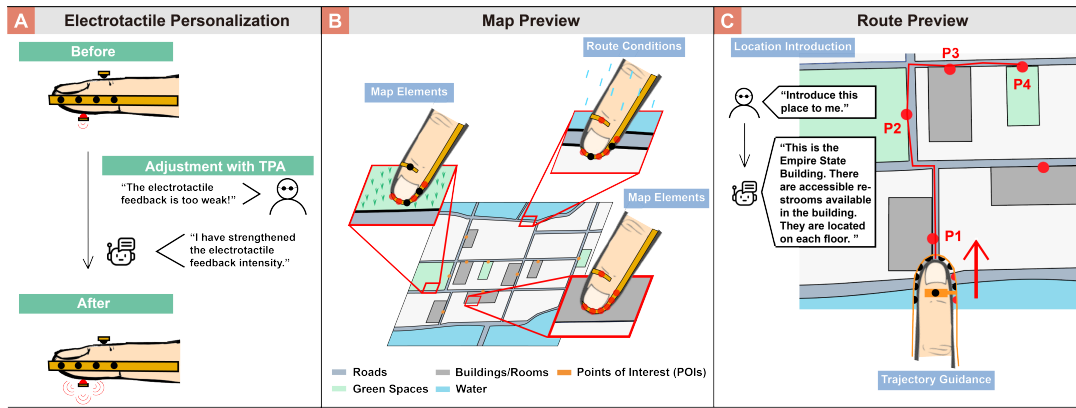


Figure 3: Overview of the Trip Preview System: A) **Electrotactile personalization:** The red cylindrical dot represents an electrode applying electrotactile feedback to the fingertip. The user communicates a weak sensation to the Trip Preview Assistant (TPA), which then adjusts the stimulation intensity. B) **Map preview:** The user explores map elements (green spaces, buildings/rooms) and route conditions (e.g., rain) through various electrotactile patterns. C) **Route preview:** The user follows electrotactile guidance along a planned trajectory (P1, P2, P3, P4) and queries the TPA for additional information (e.g., about P1). The legends for map elements are shown at the bottom.

solutions toward the vision of a continuous and personalized data interaction ecosystem for BLV users.

Achieving this vision requires optimizing electrotactile feedback through deeper exploration of its underlying mechanisms, assisted by AI. Previous work and my prior user studies showed that the effectiveness of electrotactile feedback is currently limited by factors such as variability in skin impedance across different body locations, which leads to inconsistent sensory perception [19, 25]. Traditional calibration processes are often tedious and time-consuming, hindering widespread adoption. Moreover, few studies have systematically examined the psychophysical mechanisms of electrotactile perception across diverse user groups (e.g., sighted vs. BLV individuals) and contexts (e.g., navigation vs. immersive texture rendering) [42]. A deeper understanding of how users perceive and interpret electrotactile signals is therefore critical for enhancing both system performance and user experience. These challenges motivate the following research question (RQ) for my ongoing work: **How can AI-assisted systems optimize electrotactile feedback through a deeper understanding of psychophysical mechanisms?**

To address the RQ, I plan to integrate AI to streamline the calibration and personalization of electrotactile systems in two scenarios: navigation and tactile texture rendering in immersive environments. These are the two additional applications I plan to explore as the last pieces of my dissertation. I am currently conducting research on the navigation scenario and plan to begin the immersive environment project after completing this work. In the navigation scenario, I will systematically explore the parameter space of electrotactile signals, including amplitude, frequency, and waveform, to identify effective encoding strategies for conveying spatial attributes like direction, distance, and orientation. The goal is to support BLV people’s navigation in both indoor and outdoor contexts. One possible approach is to train AI models that learn user-specific preferences and sensitivities over time, allowing the system to automatically recommend or adjust signal parameters for improved performance and comfort.

However, I am also exploring alternative solutions for this project, which I hope to discuss further during the panel to gain feedback and refine my approach. In immersive environments such as VR and AR, I will investigate how users perceive a variety of tactile textures delivered via electrotactile stimulation. Similar to the navigation context, I will explore developing AI models to generate and render context-specific textures by dynamically adjusting signal parameters, enabling scalable and adaptive tactile experiences. An open question is how to best evaluate these systems. For instance, I may conduct comparative user studies across both short-term (weeks) and long-term (months) deployments. For navigation tasks, I will assess performance metrics such as route-following accuracy and task efficiency in both indoor and outdoor environments. For immersive texture rendering, I will evaluate the perceived realism and engagement of the tactile experiences across various scenarios.

Together, these investigations will advance the fundamental understanding of electrotactile feedback and pave the way for the development of intelligent, adaptive, and user-centered haptic assistive technologies. During the Doctoral Symposium, I hope to discuss how we can better leverage AI to support the daily lives and work of BLV people through the integration of electrotactile and audio feedback. I look forward to exchanging ideas on potential future research directions, receiving feedback on my approach, and exploring ways to further refine our system to more effectively support BLV users in their daily tasks.

6 Dissertation Status and Long Term Goals

I am currently a third-year Ph.D. student in Computational Media and Art (CMA) at The Hong Kong University of Science and Technology (Guangzhou), advised by Dr. Mingming Fan and co-advised by Dr. Pan Hui. I have completed all the required coursework for my program and successfully passed the PhD Qualifying Exam (PQE), which confirmed my potential to complete the doctoral degree. I aim to finish the proposed research projects and defend

my dissertation before Summer 2026. After graduation, I plan to pursue research positions in academia, where I will continue exploring how to optimize electrotactile feedback through AI-assisted investigations of underlying psychophysical mechanisms.

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